

M KARTHIKEYAN

OT Cybersecurity Analyst | Security Assessor | AI-Enabled Threat Detection

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🌐 **Portfolio:** karthikeyan1708.github.io — *Interactive portfolio with projects, engagements & experience*

Objective

OT Cybersecurity Analyst with **1.8+ years at PricewaterhouseCoopers**, delivering high-impact security assessments, firewall reviews, and compliance audits for critical-infrastructure organizations across aviation, energy, telecom, and manufacturing. Seeking opportunities to leverage AI/ML techniques for advanced threat detection and security operations in challenging environments.

Education

B.Tech, Computer Science & Engineering (Artificial Intelligence) 2020–2024
School of AI & Computing, Amrita Vishwa Vidyapeetham, Coimbatore CGPA: 8.5 / 10
Key Coursework: Network Security, Cryptography, Machine Learning, Operating Systems, Computer Networks, Database Management

Experience

Associate – OT Cybersecurity Oct 2024 – Present
PricewaterhouseCoopers (PwC), Hyderabad 1.8+ years

OT Security Assessments & Audits

- Delivered **6 end-to-end OT cybersecurity assessments** across aviation, energy, telecom, and manufacturing sectors, encompassing network architecture review, control-system evaluation, and policy gap analysis.
- Executed on-site security audits across **4 geographically distributed industrial sites** under aggressive 3-week timelines, delivering comprehensive maturity assessments with prioritized remediation roadmaps.
- Performed in-depth **OT firewall rule analysis and network segmentation reviews**, uncovering 25+ critical misconfigurations per engagement and proposing zone-based architectures aligned with IEC 62443.
- Conducted physical and logical security reviews of telecom infrastructure across 5 tower locations, evaluating SMPS configurations, access controls, and environmental safeguards.
- Carried out **revalidation assessments** for critical OT and IT systems, performing evidence-based control analysis and maturity scoring against ISO 27001:2022 benchmarks.

PCAP Analysis & AI-Augmented Threat Detection

- Engineered Python workflows to parse and analyze **multi-terabyte PCAP datasets**, extracting 50+ protocol-level features for anomaly classification.
- Developed an **ML-based anomaly detection model** processing 500K+ daily OT sensor events with 85% precision, reducing false-positive rates by 30%.
- Built an **XGBoost risk-scoring engine** that ranked vulnerabilities by exploitability and asset criticality, cutting manual triage effort by **40%**.
- Created **NLP-based policy analysis tools** to identify compliance gaps from security documentation, achieving 90%+ extraction accuracy.

SOC Operations & Continuous Monitoring

- Providing **24/7 shift-based SOC support** for a global industrial technology provider – real-time threat monitoring, alert triage, and incident escalation using Microsoft Sentinel and Defender for IoT.
- Developed correlation rules and playbooks to improve mean-time-to-detect (MTTD) and mean-time-to-respond (MTTR) for OT-specific threat scenarios.

Project Management & Business Development

- Managed **full project lifecycle** for a multi-phase security engagement – from kickoff through final closure, including stakeholder coordination, milestone tracking, and payment reconciliation.

- Co-authored **technical proposals** for two enterprise clients in FMCG and industrial sectors, contributing threat-landscape analysis and solution architecture sections.
- Maintained comprehensive **audit documentation**, including questionnaires, working papers, evidence trackers, and executive summary reports for partner-level review.

Projects

Automatic Speech Recognition & Emotion Detection – Tamil Language *Apr–May 2024*

- Fine-tuned OpenAI Whisper with LSTM layers for emotion classification, achieving **85% accuracy** across 6 categories on a custom Tamil-speech corpus.
- Designed end-to-end data pipeline (audio augmentation, MFCC extraction, batch inference) and deployed as a containerized FastAPI service.

Communicable Disease Outbreak Prediction System *Jan–Mar 2023*

- Built RNN + SVR ensemble achieving **92% accuracy** for dengue/chikungunya forecasting over 12-month horizons using 2M+ epidemiological records.
- **Published** in Springer – *Inventive Communication and Computational Technologies* (Oct 2023).

Technical Skills

- **OT/ICS Security:** SCADA/DCS Assessment, Network Segmentation, OT Firewall Hardening, Industrial Protocols (Modbus, DNP3, OPC-UA)
- **Security Tools:** Wireshark, Claroty, Tenable, Microsoft Sentinel, ArcStrike, Defender for IoT, Nessus
- **Compliance & Frameworks:** ISO 27001:2022, IEC 62443, NIST CSF, CIS Controls, Maturity Scoring Models
- **Scripting & Automation:** Python (Advanced), PowerShell, Bash, FastAPI, Docker, Git
- **AI for Security:** ML Anomaly Detection, NLP for Policy Analysis, XGBoost Risk Scoring
- **Cloud Security:** Azure Sentinel, AWS Security Hub, Cloud-Native Monitoring

Certifications

ISO 27001:2022 Lead Implementer – Information Security Management Systems (ISMS)

Research Works

M Karthikeyan et al., “Communicable Disease Prediction Using Machine Learning and Deep Learning Algorithms,” *Inventive Communication and Computational Technologies*, pp. 979–992, Springer, Oct 2023.

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